ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

MECHATRONICS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: what mechatronics is, workshop safety and tools, mechanical systems (levers, gears and pulleys), basic electrical and pneumatic systems, sensors and actuators (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 Mechatronics combines …

A mechanical engineering, electronics, control and computing.
B only mechanical engineering.
C only computer programming.
D only electrical wiring.

1.1.2 A seesaw is an example of a …

A second-class lever. B first-class lever. C third-class lever. D pulley.

1.1.3 When a small driver gear turns a large driven gear, the driven gear turns …

A faster, with more torque.
B faster, with less torque.
C at the same speed.
D slower, with more torque.

1.1.4 The unit of force is the …

A newton. B joule. C watt. D pascal.

1.1.5 A pneumatic system uses …

A oil under pressure.
B electricity only.
C compressed air.
D water only.

1.1.6 A device that converts an electrical signal into straight-line movement is a …

A potentiometer.
B thermistor.
C linear actuator.
D capacitor.

1.1.7 Which tool is used to measure a shaft's diameter accurately to 0,01 mm?

A Micrometer
B Steel rule
C Tape measure
D Try square

1.1.8 A sensor whose resistance changes with temperature is a …

A LDR. B reed switch. C solenoid. D thermistor.

1.1.9 Two gears of equal size that mesh will turn …

A in the same direction at different speeds. B in opposite directions at the same speed.
C in the same direction at the same speed. D in opposite directions at different speeds.

1.1.10 A washing machine, an automatic gate and a car's ABS brakes are all examples of …

A purely mechanical systems. B hydraulic systems only.
C hand tools. D mechatronic systems.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Mechanical advantage = load ÷ effort.

1.2.2 A wheelbarrow is a third-class lever.

1.2.3 A belt drive can slip if it is too loose.

1.2.4 An idler gear changes the gear ratio.

1.2.5 A single-acting cylinder uses a spring to return the piston.

1.2.6 A limit switch can tell a controller that a door is fully open.

1.2.7 Work done = force × distance.

1.2.8 Loose clothing is safe to wear when working with rotating machines.

1.2.9 A chain and sprocket drive is used on a bicycle.

1.2.10 A relay allows a small current to switch a larger current.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	The point about which a lever turns	A	pulley
1.3.2	Turning effect of a force	B	worm and wheel
1.3.3	Converts rotary motion into reciprocating (back-and-forth) motion	C	directional control valve
1.3.4	Converts rotary motion into straight-line motion with a toothed bar	D	crank and slider
1.3.5	Wheel with a groove for a rope or belt	E	cam
1.3.6	Supplies compressed air to a pneumatic system	F	fulcrum
1.3.7	Controls the direction of air flow to a cylinder	G	rack and pinion
1.3.8	Gear that allows rotation at 90° to the input, used to reduce speed greatly	H	compressor
1.3.9	Reduces friction between a shaft and its housing	I	torque (moment)
1.3.10	Shaped part that turns to give a follower a special movement	J	bearing

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MECHATRONICS, SAFETY AND TOOLS

2.1 Explain what a mechatronic system is, and describe how an automatic sliding door works as a mechatronic system (input, process, output). (6)

2.2 Give FOUR safety rules for working in a mechatronics workshop. (4)

2.3 Name the tool you would use for EACH task: (a) cutting a thin metal sheet; (b) tightening a hexagonal nut; (c) stripping insulation from a wire; (d) checking that a surface is level. (4)

2.4 Name the THREE classes of levers and give an everyday example of each. (6)

[20]

QUESTION 3: MECHANICAL SYSTEMS

3.1 A crowbar is used to lift a rock. The effort of 150 N is applied 1,2 m from the fulcrum and the rock is 0,3 m from the fulcrum on the other side.

3.1.1 Calculate the load (weight of the rock) that can be lifted. (3)

3.1.2 Calculate the mechanical advantage. (2)

3.2 A motor gear with 20 teeth drives a gear with 60 teeth. The motor turns at 300 r/min.

3.2.1 Calculate the gear ratio. (2)

3.2.2 Calculate the speed of the driven gear. (2)

3.2.3 Explain what happens to the torque at the driven gear. (2)

3.2.4 What must be added so that the driven gear turns in the same direction as the motor gear? (1)

3.3 A pulley system with 4 supporting ropes lifts a load of 800 N. Calculate the ideal effort needed. (3)

3.4 A robot lifts a box with a force of 50 N through a height of 1,5 m in 3 seconds. Calculate (a) the work done; (b) the power. (4)

3.5 Name TWO ways to reduce friction in moving machine parts. (2)

[21]

QUESTION 4: ELECTRICAL AND PNEUMATIC SYSTEMS

4.1 Draw a simple circuit in which a battery, a switch and a fuse control a DC motor. Use correct symbols. (5)

4.2 A conveyor motor uses 24 V and 2 A. Calculate its power. (2)

4.3 Give THREE advantages of pneumatic systems compared with hydraulic systems. (3)

4.4 Explain how a single-acting cylinder controlled by a 3/2 push-button valve works. (4)

[14]

QUESTION 5: SENSORS AND ACTUATORS

5.1 Explain the difference between a sensor and an actuator, and give TWO examples of each. (6)

5.2 Suggest a suitable sensor for EACH application: (a) detecting when a metal part is on a conveyor without touching it; (b) switching on garden lights when it gets dark; (c) detecting that a window is open in an alarm system. (3)

5.3 An automatic chicken-coop door must open at sunrise and close at sunset. Name the input, controller and output devices needed. (3)

5.4 Explain why mechatronic systems need regular maintenance and name TWO maintenance tasks. (3)

[15]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER